

Map Mapping Markers

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1. Course Content

[!IMPORTANT]

- This section is a comprehensive course that demonstrates and explains the use and operation steps of the sand table map. For detailed parameter debugging and source code explanation, refer to the prerequisite course: [05-OpenClaw SLAM Mapping Navigation]
- Mark map mapping for target locations that need to be used in the sand table map.

2. Preparation

2.1 Map Preparation

- First, complete the construction of the sand table map and pose map according to the [Build Sand Table Map] course.

2.2 Start the MicroROS Chassis Agent

- Skip if already started

```
sh start_agent.sh
```

2.3 Start the Road Network Container

```
roadnet start
```

2.4 Start Odometry, tf, Robotic Arm Assistant, Camera, Aritag Detection Nodes

```
ros2 launch m3pro_bringup car_base.launch.py relocation:=True
```

2.5 Start the MCP Service

```
ros2 launch m3pro_bringup mcp_service.launch.py
```

2.6 Start Road Network Navigation

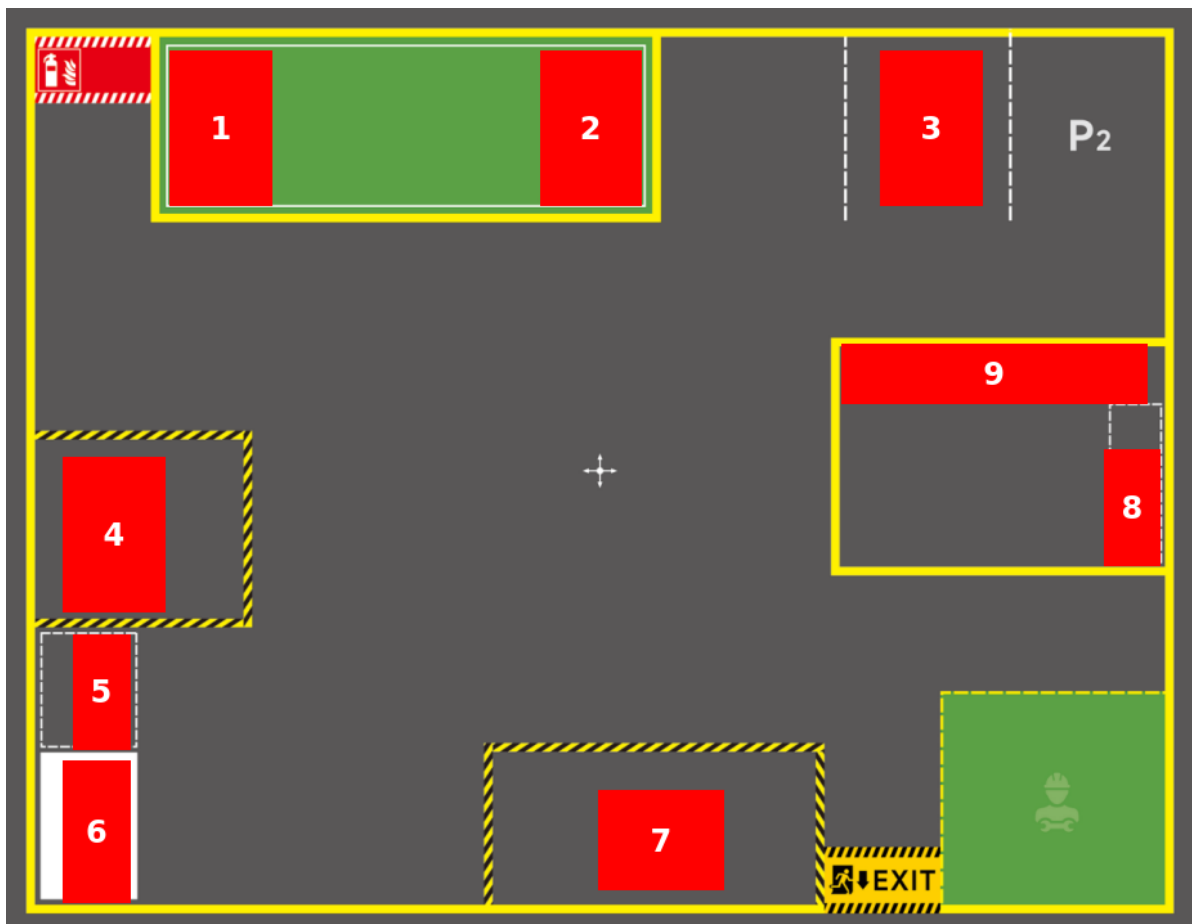
- Start inside the roadnet container

```
ros2 launch road_net_route roadnet_nav2.launch.py  
rviz_config_file:=/root/volumes/code_ws/src/road_net_route/rviz/nav2_relocaton.r  
viz
```

3. Record Map Mapping

- The layout and area naming of the map site are arbitrary and can be modified according to needs. The following reference area naming cases refer to:

1. Shelf 1
2. Shelf 2
3. Parking lot 1
4. Cargo warehouse
5. Cargo sorting area
6. Shelf 3
7. Shipping area
8. Garbage sorting area
9. Trash bin locations: including kitchen waste, recyclable waste, other waste, hazardous waste



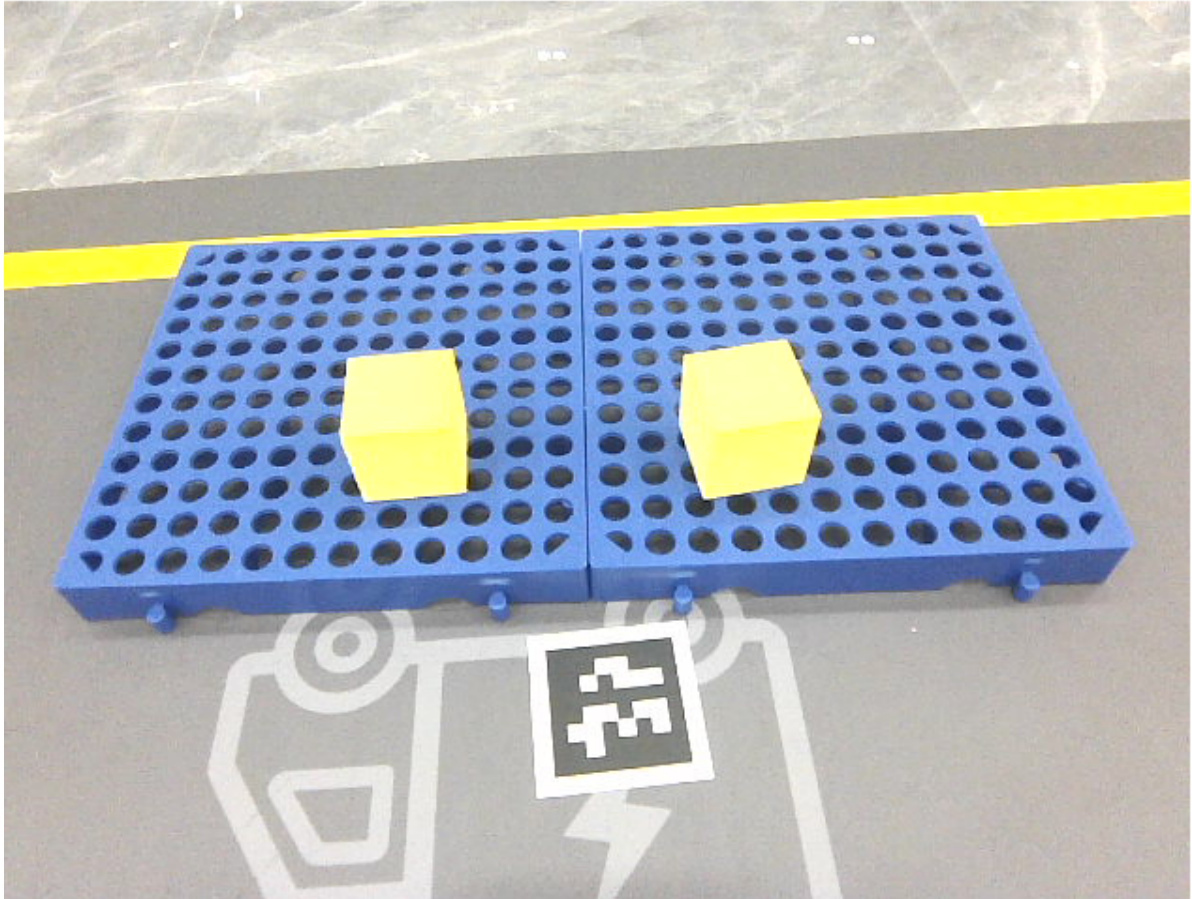
- Move the robot to the location that needs to be recorded, and use the robot_control CLI tool to record the map mapping.

```
robot_control call-tool RecordMapLocation --name xxx --symbol x
```

Important: Before recording a location, always check whether the laser contour matches the environmental edge features. If there is a deviation, use the **2D Estimate** tool for manual relocation before recording the position.

Recording example:

- Record the shipping area. Move the robot to the location near the shipping area that needs to be recorded.
- Ensure that the cargo in the area is fully visible in the field of view.



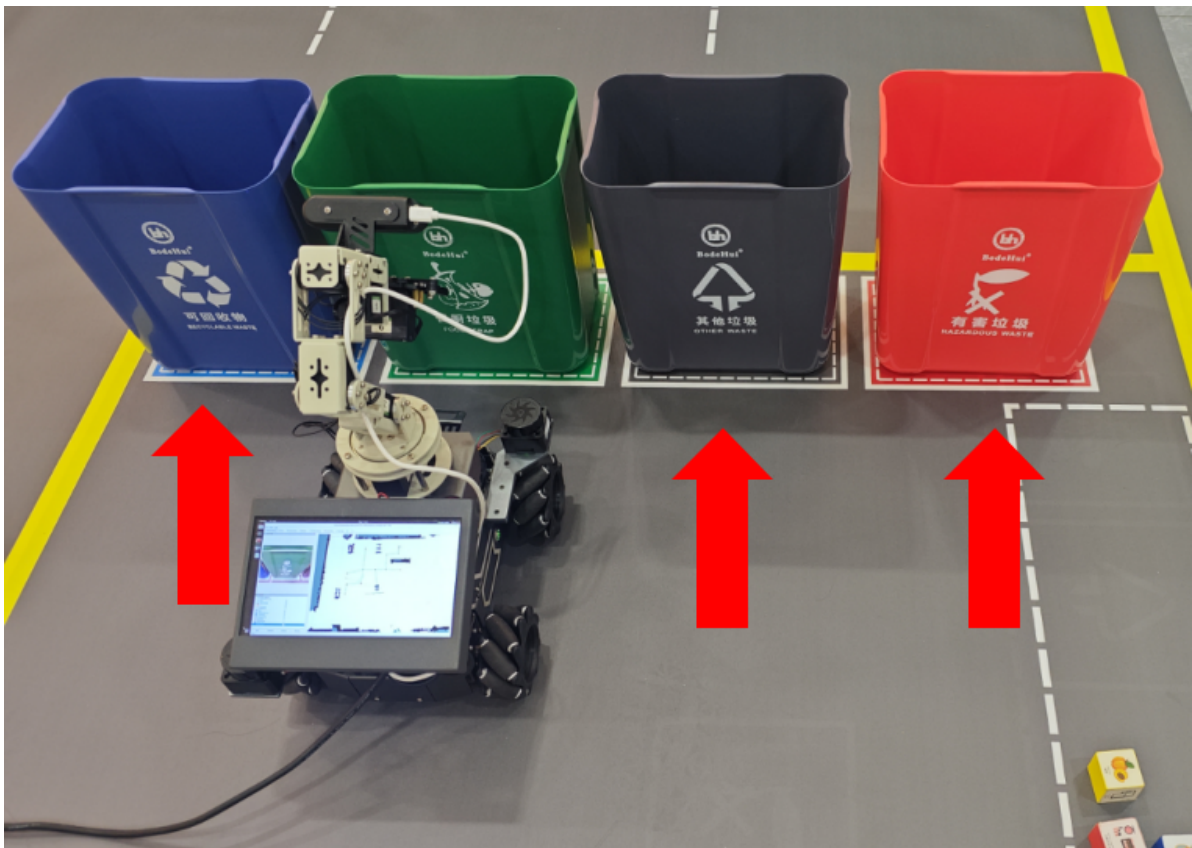
- Use the robot_control CLI tool to record the position of the robot's **base_link** in the **map** coordinate system.

```
robot_control call-tool RecordMapLocation --name 发货区 --symbol B
```

- --name specifies the name of the recording point
- --symbol specifies the symbol of the recording point, used by OpenClaw for identification and differentiation.

```
jetson@yahboom:~$ robot_control call-tool RecordMapLocation --name 发货区 --symbol B
{
  "execution_result": "success",
  "reason": ""
}
```

- The positions of different types of trash bins need to be recorded separately. Place the robot in front of the trash bin to be recorded, then perform the recording.



- Check the map mapping file `map_mapping.yaml`. The file path is:

```
$HOME/M3Pro_ws/src/m3pro_bringup/config/map_mapping.yaml
```

- Check whether all required locations have been recorded.

```
jetson@yahboom:~$ cat $HOME/M3Pro_ws/src/m3pro_bringup/config/map_mapping.yaml
A:
  name: OtherWaste
  position:
    x: 1.033
    y: 0.483
    z: 0.0
  orientation:
    x: 0.0
    y: 0.0
    z: 1.0
    w: 0.008
B:
  name: 发货区
  position:
    x: 2.078
    y: -0.96
    z: 0.0
  orientation:
    x: 0.0
    y: 0.0
    z: -0.025
    w: 1.0
D:
  name: RecyclableWaste
  position:
    x: 1.09
    y: -0.259
    z: 0.0
  orientation:
    x: 0.0
    y: 0.0
    z: 1.0
    w: 0.009
C:
```